Hamro Life Bank ™

Design and Requirement Document

For: **Rumsan**
By: **Aishwarya Maharjan**
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Introduction

Nepal faces a persistent challenge in blood supply chain management. Hospitals and patients in need of blood often encounter delays due to fragmented donor databases, manual coordination, and a lack of real-time blood availability information. Informal communication through phone calls and messaging apps leads to errors, missed donors, and critical delays in emergencies.

Hamro Life Bank (HLB) is a centralized, digital blood bank management system designed to address these systemic gaps. Built as a secure, web-based and mobile-responsive platform, HLB streamlines the end-to-end blood request lifecycle — from intake and donor matching to fulfillment and analytics — while serving call operators, administrators, volunteers, and donors through role-based interfaces.

This document defines the functional and non-functional requirements, system architecture, user roles, use cases, API endpoints, and data models required to design and build the Hamro Life Bank platform.

Objectives

- Centralize blood request and donor management on a single digital platform.
- Enable real-time blood inventory tracking across blood banks inside and outside the Kathmandu Valley.
- Implement intelligent donor matching based on blood type, location, and availability.
- Automate notifications, follow-ups, and status updates to reduce manual coordination.
- Provide management with analytics on request resolution, donor response, and system performance.
- Support multi-channel communication (Phone, SMS, WhatsApp integration roadmap).

Scope

The current version of Hamro Life Bank covers the following functional modules:

- Blood request intake, tracking, and resolution (Call Operator & Admin flows).
- Donor database management with lifecycle tracking (Active, Pledged, Blacklisted).
- Blood bank inventory management segmented by location (Inside Valley / Outside Valley).
- Feedback collection from both patients and donors.
- Role-based access control for Admin, Call Operator, Volunteer, and Donor roles.
- Analytics and reporting dashboard for performance monitoring.

Technical scope is constrained to: OAuth-based Google authentication, D1 (SQLite) as the primary database, REST API architecture, and a React-based web frontend with mobile-responsive design.

Document Version

Version	Changes	Date	Author	Status
1.0	Initial version	04/29/2026	Aishwarya	Released

Development Schedule

Role Assignments:

- IT — Development Team
- ST — Support / QA Team
- EU — End Users (Operators, Admins)

Development Stage	Apr 202026	Apr 272026	May 042026	May 082026	May 122026	May 152026
Phase 1: Planning & Requirement	IT/ST					
Phase 2: UI/UX & DB Design		IT	IT			
Phase 3: Core Module Development			IT	IT	IT	
Phase 4: Integration & Testing				IT/ST	IT/ST	
Phase 5: UAT & Validation					EU	
Phase 6: Deployment						ST/IT
Phase 7: Maintenance						IT

Work Flows

High-Level Process Flow

End User Relationship

User Story

Project Technology Requirements

- Turbo Monorepo
- pnpm
- next js
- tailwindCss
- axios
- tanstack query
- zustand
- zod validation
- Hono
- D1 Database
- SQLite
- Drizzel ORM
- typescript
- Node.js
- LLM suggest best one for this task (Gemini / GPT / Claude)

Non-Functional Requirements

Security and Authentication

- The application must be a secure, web-based platform accessible over HTTPS.
- Authentication must support OAuth-based Google login
- The system must enforce Role-Based Access Control (RBAC) ensuring users only access features permitted for their role.
- Passwords (for non-OAuth users) must be reset every 6 months as per policy.

Performance and Availability

- The system must be mobile-responsive to allow operators and donors to use it on mobile devices.
- Blood request status updates must be reflected in real-time (under 3 seconds).
- The platform must deliver automated notifications and follow-up reminders without manual intervention.
- System uptime target: 99.5% during operating hours (9:00 AM – 6:00 PM, Asia/Kathmandu timezone).

User Roles

User Role	Permissions & Responsibilities
Admin/Operator	• Can login via OAuth-based Google authentication. • Can manage users and roles. • Can maintain hospital and blood bank master lists. • Can view all analytics and reports. • Can manage and resolve operational tasks and reported facility issues. • Can update the status of service requests in real-time. • Can see the user roles of other employees. • Coordinates digital management to improve workplace efficiency
Volunteer	• Can view and search blood requests. • Can view donor list (read-only). • Cannot create, update, or delete requests or donor records.

Requirements

	General - Requirements	MoSCoW	Version
GR-001	The system must be a secure, web-based platform	M	1
GR-002	The application must be mobile-responsive.	M	1
GR-003	The system must centralize blood requests and donor management workflows.	S	1
GR-004	The system must automatically redirect authenticated users from the homepage directly to their respective Dashboard.	M	1
GR-005	The system should push notification to the user when any operation is performed.	S	1
GR-006	The system must auto-capitalize data inputs for consistency.	M	1

	Security - Requirements	MoSCoW	Version
SR-001	The system must handle authentication via OAuth-based login using Google.	M	1
SR-002	The system must strictly enforce Role-Based Access Control (RBAC).	M	1
SR-003	The system must store all user data and records securely in a PostgresSQL database	M	1
SR-004	The system should maintain data integrity for all blood requests and donor records.	S	1
SR-005	Passwords for non-OAuth accounts must be reset every 6 months.	M	1

	Blood Request Requirements	MoSCoW	Version
BRR-001	The system must allow operators to create blood requests with full patient and requester details.	M	1
BRR-002	The system must assign a unique ID to every blood request for tracking.	M	1
BRR-003	New requests must appear at the top of the request list.	M	1
BRR-004	The system must support backdated request entry.	M	1
BRR-005	The system must provide real-time status: New, In Progress, Managed, UnManaged & Verified.	M	1
BRR-006	Blood Required Date must default to today's date.	M	1
BRR-007	The system must send automated follow-up reminder 1 hour after blood is marked Managed.	M	1
BRR-008	If blood is in stock, system must display eligible hospitals and allow reservation.	M	1
BRR-009	Hospitals should support auto-suggestion with location autofill.	S	1

	Donor Requirements	MoSCoW	Version
DR-001	The system must allow operators to search donors by blood type, location, and availability.	M	1
DR-002	The system must support donor categories: Active, Pledged, Blocklisted.	M	1
DR-003	Donors marked Do Not Call or Dormant must be auto-moved to Blocklisted.	M	1
DR-004	The rating system must be mandatory and provide a 10-second undo window after submission.	M	1
DR-005	The system must show Last Contacted Date, Status, and Remarks in the donor list view.	M	1
DR-006	Website pledges should go to Pledged status; operator verification moves them to Active.	S	1
DR-007	Entering a last donated date should auto-increment total donation count.	S	1
DR-008	Communication types should be restricted to Phone Call and SMS.	S	1
DR-009	Remarks and Feedback fields must be merged into a unified Notes/Comments field.	S	1

	Blood Bank & Inventory Requirements	MoSCoW	Version
BBIR-001	The system must maintain a blood inventory segmented by Inside Valley and Outside Valley.	M	1
BBIR-002	Blood bank and hospital lists must be editable from the dashboard.	M	1
BBIR-003	The system must display real-time blood availability before donor matching.	M	1
BBIR-004	Managed hospitals should auto-suggest based on prior request history.	S	1

	Feedback Requirements	MoSCoW	Version
FR-001	The system must collect patient feedback on resolved blood requests.	M	1
FR-002	The system must provide a separate Donor Feedback tab linked to request actions.	M	1
FR-003	Donor feedback must handle cases where the assigned donor did NOT donate.	M	1
FR-004	Patient Feedback form should remove Contacted/Not Contacted fields.	S	1

	Reporting Requirements	MoSCoW	Version
RR-001	Admins must be able to view request resolution and response time reports.	M	1
RR-002	Charts and reports must be exportable to Excel.	M	1
RR-003	Reports must clearly display totals and summary statistics.	M	1
RR-004	The system must track donor conversion rates.	M	1

	Admin Requirements	MoSCoW	
AR-001	Admins must manage users with roles: Admin, Volunteer.	M	1
AR-002	Admins must be able to deactivate user accounts.	M	1
AR-003	Admins should be able to configure SLA hours per priority level.	M	1
AR-004	Admins must be able to manage system notification settings.	S	1

	Volunteer Requirements	MoSCoW	Version
VR-001	Volunteer must be able to access to Donor and Patient list. (read only)	M	1
VR-002	The system must show blood bank availability. (read only)	S	1
VR-003	The system must show active blood request. (read only)	S	1

Use Cases